

TaskFlow Product Health Monitor

Multi-agent ElevenLabs voice workflow — 05-elevenlabs-producthealth

Problem

TaskFlow's product-health signals user feedback, support tickets, product metrics, but are scattered across domains. A PM asking a spoken question shouldn't need to know which dashboard or team owns the answer, and a single do-everything prompt risks scope-bleed between domains.

Approach & Key Decisions

- Router-only Greeter: explicitly forbidden from answering data questions itself; keeps every answer coming from exactly one scoped specialist.
- One KB file per specialist: a harder guarantee against cross-contamination than a prompt instruction alone.
- LLM-condition edges (natural language): some determinism for a natural voice UX.
- Bidirectional cross-routing edges let a caller change topic mid-call without restarting at the Greeter.

What It Does

- Greeter classifies intent and hands off to one of 3 specialists (never answers itself).
- User Feedback / Support Ticket / Product Metrics analysts, each scoped to one KB file.
- Cross-routing edges let any specialist hand off directly to another mid-conversation.
- Each specialist redirects politely on out-of-scope questions rather than guessing.
- 12 total routing conditions: 3 forward, 3 to End, 6 cross-routing (3 bidirectional edges).

Results & Learnings

- Node-level KB restriction is a stronger scoping mechanism than relying on prompt instructions alone.
- A single bidirectional edge object per specialist pair kept the graph readable (3 objects instead of 6).
- This build doubled as a lesson in backup discipline: an earlier version of this workflow was lost to an accidental canvas clear, and the platform's own version history only captured a pre-build “first publish” checkpoint: codebase/agent-workflow-config.json exists specifically as an external backup.

Tech Stack

Component	Value
Platform	ElevenLabs Agents (Agent Workflows)
LLM	Gemini 2.5 Flash
ASR	Scribe (scribe_realtime), PCM 16kHz
TTS	eleven_flash_v2
Orchestration	Node/edge workflow graph, LLM-condition routing
Knowledge base	Per-node scoped .docx (RAG, direct attachments)